Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/04/2023
PI Name:	Yun, Sanghee Ph.D.		
Project Title:	Exposure of Male and Female Mice to 33-Particle Galactic Cosmic Radiation (GCR): Negative Impact on Translationally-relevant Hippocampal-based Cognition and Prevention of GCR-induced Deficits by Acute Treatment with the Anti-inflammatory CDDO-EA		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4318	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	10/01/2021	**End Date:**	09/30/2023
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per S. Mack/JSC and NSSC information (Ed., 5/22/23) NOTE: End date changed to 03/31/2023 per NSSC information (Ed., 10/12/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Eisch, Amelia Ph.D. (Children's Hospital of Philadelphia)		
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Task Description:	Background: Galactic Cosmic Radiation (GCR) is an unavoidable factor in deep space that may affect astronaut health and mission success. To date, ground-based studies on the cognitive impact of GCR have exposed rodents to a single high-energy particle or a combination of a few particles. It is unknown how a complex, multiple-ion GCR spectrum -- as will be encountered during a Mars mission -- influences rodent cognition. It is also important to identify countermeasures that can reduce risk to crew health or performance. Preclinical and clinical studies suggest the anti-inflammatory compound CDDO-EA mediates its therapeutic effects in combating lung disease by upregulating the key antioxidant transcription factor Nrf2. Notably, in mouse models of neurological disorders, CDDO-EA ameliorates striatal- and hippocampal-dependent cognitive and cellular deficits. Thus, it is timely and urgent to assess if a complex GCR spectrum decreases rodent cognition and if CDDO-EA mitigates such changes. Hypothesis: Exposure to a complex GCR mixture will decrease translationally-relevant cognitive function and negatively impact associated cellular and molecular processes in female and male mice, and CDDO-EA will mitigate these changes. Aim 1. We will determine the effects of (1a) 33-particle GCR (33-GCR) on hippocampal dentate gyrus (DG) function (pattern separation) and prefrontal cortex (PFC) functions (cognitive flexibility, extinction of stimulus-response) in 6-month-old mature male C57BL/6J mice and (1b) CDDO-EA on 33-GCR-induced cognitive changes. Aim 2. We will also define DG neurogenesis and DG and PFC inflammation in female and male mice exposed to 33-GCR or Sham with and without CDDO-EA. Methods: Mature C57BL/6J male and female mice will receive CDDO-EA (400 mg/kg, n=80) or Control Chow (Control; n=80) for 5 days (3 days before, during, and 1 day after 33-GCR). Mice will receive either Sham or whole-body 33-GCR (~73% 1H, 18% 4He, 2.4% 12C, 3.1% 16O, 1.6% 28Si, 0.9% 48Ti, 0.8% 56Fe; 750mGy) with the 4 groups being Control/Sham, Control/33-GCR, CDDO-EA/Sham, CDDO-EA/33-GCR (n=16/group). To complement pilot data with female mice, in Aim 1, 2-mon post-irradiation male mice will undergo general touchscreen training, location discrimination (LD), and acquisition and extinction of stimulus-response. In Aim 2, brains will be collected from male and female mice at three timepoints post-33-GCR: 3 days, 2-mon, and 11-mon. 2- and 11-mon groups will be given bromodeoxyuridine (BrdU, i.p. 180 mg/kg) 2 hrs prior to perfusion and indices of neurogenesis and inflammation will be quantified via unbiased stereology. In the 3-day group, protein expression of Nrf2/ARE signaling targets will be quantified in the DG and PFC. Aim 1 deliverables include measures of LD (session length, # trials, % correct, etc.) and acquisition and extinction (session length, # responses, # omissions, days to completion) in male mice, and the impact of 33-GCR vs. Sham and CDDO-EA vs. Control Chow on these measures. Aim 2 deliverables include indices of DG neurogenesis (# of Ki67-, DCX-, BrdU-immunoreactive [+] cells) and DG and PFC inflammation (# of Iba-1+, Glial fibrillary acidic protein (GFAP) + cells)), and protein expression of signaling mediators downstream from nuclear factor erythroid 2–related factor 2 (Nrf2)/antioxidant response element (ARE). Significance: It is imperative to understand how a mission-relevant GCR mixture influences cognition, and determination of GCR-induced cellular and molecular changes enables identification of appropriate countermeasures. The proposed work in male and female mice will complement our already-completed cognitive study in female mice (manuscript in preparation). Given the similarity between cognitive function in mice and humans, the resulting data will allow more accurate estimation of space-radiation risks to both male and female astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is imperative to understand how a mission-relevant GCR mixture influences cognition, and determination of GCR-induced cellular and molecular changes enables identification of appropriate countermeasures. Our work in male and female mice will complement our already-completed cognitive study in female mice (manuscript in preparation). Given the similarity between cognitive function in mice and humans, the resulting data will allow more accurate estimation of space-radiation risks to both male and female astronauts. Our study will elucidate what the acute and late CNS risk are at the molecular, cellular, and behavioral levels for astronauts exposed to GCR.
Task Progress:	Dr. Sanghee Yun, a Principal Investigator, provides the day-to-day supervision of the team. Dr. Eisch, as Co-investigator, and Dr. Fred Kiffer, postdoctoral fellow, advise Dr. Yun based on their previous NASA runs and running NSRL 22A together with Dr. Yun at Brookhaven National Lab (BNL). At Children's Hospital of Philadelphia (CHOP), the current research staff that spearheads the spontaneous location recognition (SLR), touchscreen and cellular experiments include postdoctoral fellow Dr. Fred Kiffer and University of Pennsylvania (PENN) work-study students Harley Haas, Grace Bancroft, and Amishi Mahajan along with summer intern program students, Stephanie Uroda, Hannah An, Elise Walle-Fridman (PENN), and Chris DeSantis (Yale U.). Experiments are completed at CHOP on female and male mice from NSRL 22A (supported by Eisch's Institutional Animal Care and Use Committee/IACUC protocols from BNL and CHOP). Our lab has made progress on our NASA-funded research as described below: NASA Human Exploration Research Opportunities (HERO) (PI: Yun)-funded publications (4): Tran FH, Spears SL, Ahn KJ, Eisch AJ, Yun S. Does chronic systemic injection of the DREADD agonists clozapine-N-oxide or Compound 21 change behavior relevant to locomotion, exploration, anxiety, and depression in male non-DREADD-expressing mice? Neurosci Lett. Elsevier. 2020 Nov. Clark LR, Yun S, Acquah NK, Kumar PL, Metheny HE, Paixao RCC, Cohen AS, Eisch AJ. Mild traumatic brain injury induces transient, sequential increases in proliferation, neuroblasts/immature neurons, and cell survival: A TimeCourse study in the male mouse dentate gyrus. Front Neurosci. 2020 Dec. Soler I*, Yun S*+, Reynolds RP, Whoolery CW, Tran FH, Kumar PL, Rong Y, DeSalle MJ, Gibson AD, Stowe AM, Kiffer FC, Eisch AJ+. *Co-first authors, +Co-corresponding authors. Multi-domain touchscreen-based cognitive assessment of C57BL/6J female mice shows whole-body exposure to 56Fe particle space radiation in maturity improves discrimination learning yet impairs stimulus-response rule-based habit learning. Front Behav Neurosci. 2021 Oct 11;15:722780. PMCID: PMC8543003. Kiffer FC, Luitel K, Tran FH, Patel RA, Guzman CS, Soler I, Xiao R, Shay JW, Yun S, Eisch AJ. Effects of a 33-ion

	sequential beam galactic cosmic ray analog on male mouse behavior and evaluation of CDDO-EA as a radiation countermeasure. Behav Brain Res. 2022;419:113677. NASA HERO (PI Yun) additional project-related progress: We examined the impact of exposure to 33-particle galactic cosmic radiation (GCR) on cognitive performance in fully mature male and female mice and the mitigation of 2-cyano-3,12-dioxoolean-1,9(11)-dien-28-oic acid (CDDO), an antioxidant provided to the mice around the time of irradiation. We purchased male (n=124) and female (n=48) wildtype (C57BL/6J) mice for NSRL 22A (April 20, 2022). The variables for this experiment were irradiation (Sham and 33-GCR) and antioxidant (Veh or CDDO-EA), resulting in four groups of mice/sex (Sham/Veh, 33-GCR/Veh, Sham/CDDO-EA, 33-GCR/CDDO-EA; n=12/group/sex). After BNL, the mice were returned to CHOP. Female and male mice were then assessed for: monthly weight checks, home cage locomotion activity monitoring (18hrs), spontaneous location recognition test (a pattern separation test), training and testing on an automated touchscreen platform (Pavlovian learning) and elevated plus maze (a test for anxiety). We are currently analyzing all behavior data, including performance on the pattern separation test, touchscreen-based Pavlovian learning tests, and anxiety tests. Brains from all mice were collected and will be processed for dependent measures based on the behavioral results. Also, we run a spontaneous location recognition test (a pattern separation test) with fiber photometry recording to assess the impact of 33-GCR and CDDO on the dentate gyrus (DG) granular cell activity in behaving animals and currently analyze the data. We have shared these data each year at NASA Human Research Program (HRP) and Radiation Research in order to receive feedback on our approach and interpretations. The resulting data are the focus of several manuscripts we have in preparation/revision, and Dr. Yun (PI of this HERO grant) is first and/or senior author on each manuscript.
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